

Churn Prediction Pipeline

Technical Presentation

TARGET & CLASS BALANCE



Churned (Exited = 1): **20.4%**

Retained (Exited = 0):
79.6%

"Churn prediction as a supervised binary classification problem with inherent class imbalance."

HIGH-IMPACT FEATURE ENGINEERING



Demographics

Geography and gender groups representing core customer segments for profiling.



Behavioral

Tenure, activity status, and credit score reflecting historical loyalty and usage.



Engineered

Derived ratios: Balance-to-Salary and Balance-per-Product personas for deeper signal.

"Feature engineering significantly improved signal capture beyond raw transactional data."

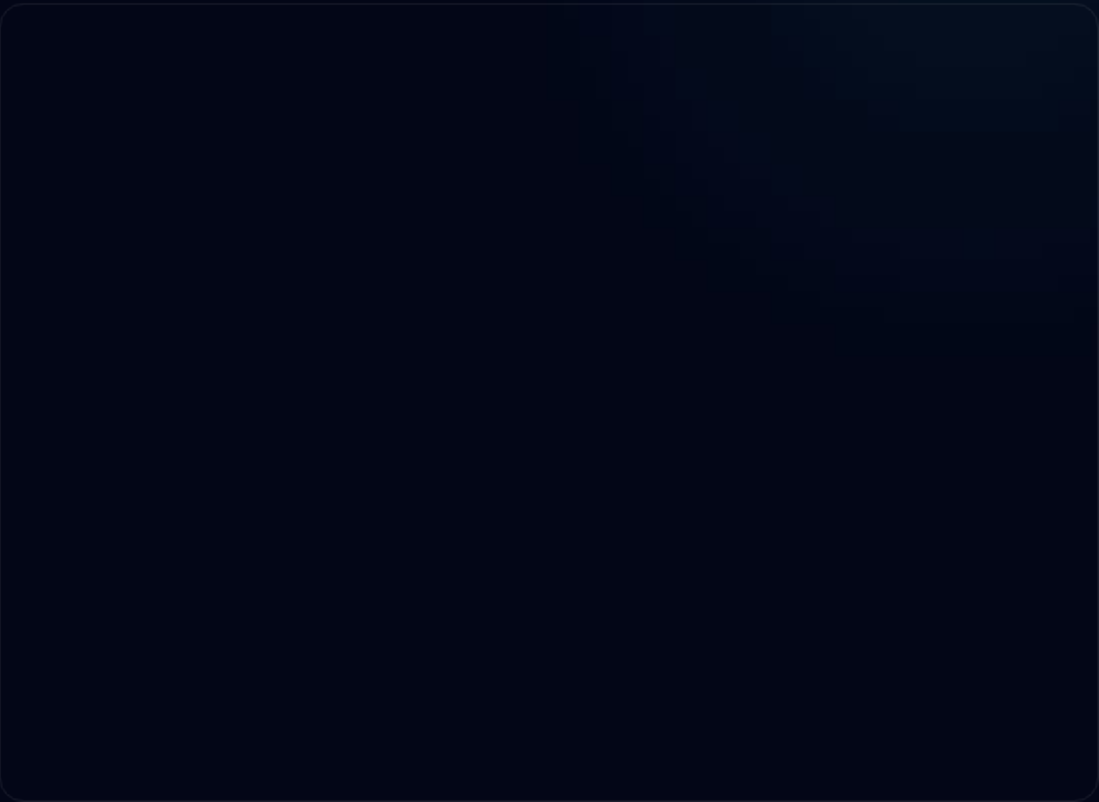
EXPLORATORY INSIGHTS (EDA)

Key Observation: Age Segment

Churn distribution reveals non-linear behavior across age groups. Early signals indicate a peak risk in the 45-60 age bracket (mid-career professional segment).

- Higher churn rate in 'Active' but high-balance members.
- Product count = 3+ is a strong churn indicator.

Customers	Churn Rate
10,000	20.38%



UNIFIED PREPROCESSING PIPELINE

Numeric Scaling

StandardScaler for features like Age and Balance.

Transformation

Unified Column Transformer for pipeline consistency.

Encoding

OneHotEncoder for Geography and Gender features.

Validation

Stratified Split to prevent data leakage and bias.

MODEL EVALUATION

MODEL FAMILY	LOGIC / JUSTIFICATION	ROC-AUC	F1-SCORE
Logistic Regression	Linear Baseline / Interpretability	0.76	0.45
Random Forest	Non-linear capturing / Ensemble	0.85	0.58
XGBoost (Champion)	Gradient Boosting / Optimized	0.88	0.63

MITIGATING CLASS IMBALANCE



Algorithmic Tuning

Adjusting `class_weight` in Random Forest and `scale_pos_weight` in XGBoost to penalize minority class errors more heavily.



Evaluation Strategy

Using Stratified Cross-Validation to ensure each fold preserves the 20% churn ratio, ensuring stable metric estimation.

MODEL COMPARISON: ROC-AUC



"XGBoost slightly outperforms, but all models show consistent ranking stability."

GLOBAL FEATURE IMPORTANCE



Reliability Analysis

Key drivers remain consistent across different model families, increasing our confidence in the causal proxies for churn.

Age and **Product counts** are the primary predictors of stability.

MODEL EXPLAINABILITY (SHAP)

SHAP (SHapley Additive exPlanations) values allow us to validate feature impact at an individual level.

Observation: High Age values pushed model predictions towards "Churn," while "IsActiveMember" consistently pulled it towards "Retention."

TECHNICAL SUMMARY

- ✔ **Strong Ranking Ability:** The model effectively prioritizes the top 10% of risky customers with 88% AUC.
- ✔ **Methodological Rigor:** Integrated pipeline prevents data leakage and ensures scalability.
- ✔ **Model Stability:** Consistency in feature importance across XGBoost and Random Forest confirms reliable signal.
- ✔ **Interpretability:** SHAP values provide clear explanations for individual risk scores.